Engineering Mechanics 1 · Unit_02 · UNIT 2

DRAFT — UE26ME142A · Unit_02 · UNIT 2 · UE26ME142A Unit 2 Class 18 AVSummary

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

UE26ME142A Unit 2 Class 18 AVSummary

1 UE26ME142A Unit 2 Class 18 AVSummary

Study focus: UE26ME142A_Unit 2_Class 18_AVSummary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000074 · EV0000075

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

My name is Dr.

Key details

- Isviving Tish from Civil Engineering under the Department of Science and Humanities, PS University.
- In the last class, that is class number twenty one, we had discussed two problems on equilibrium.
- Today we will be discussing some more problems.
- So first one that I have taken as 3 bar 27.
- The statement of the problem is find the angle of tilt theta with the horizontal so that the contact force at B will be one half that at A for the smooth cylinder.
- The statement what we could identify as that we need to find out the angle theta for the condition that the contact force acting at B is equal to one half of that acting at A.
- On this V block, they have kept a.
- If the mass of the cylinder is M, then the sulphate of the cylinder would be mass times the accuration due to gravity, which will be acting at the center.
- The contact point of the cylinder on the V groove is at two point, that is A, that is this point, and B that is this point.
- The self weight will be acting downwards.
- That has to be balanced by the reaction.
- You can know that this line is the tangent to the cylinder.
- Because that would be only at one contact point, and that would be always going to be tangent to the cylinder.

- So the reaction, if it is normal to this, then that means it would be passing through the radial line.
- Which will also pass through the center of the cylinder.
- This reaction at B is going to be normal to this line or this groove, which is also tangential to the cylinder.
- Which will also pass through the center of the cylinder.
- So you will have three different forces acting because there are only two contact points where the cylinder is touching the V group, that is at A and B.
- Sulfate of the cylinder that is going to be acting.
- Next, to know the angle at which the reaction at A and B is acting, we have to identify this, which is 45 degrees.
- The angle made by the group, V Group, which is passing through A.
- With respect to the horizontal, we'll be at an angle of 45 degrees minus this angle theta, which is 45 minus theta.
- Whereas here, if I you find out the angle made by the group passing through B with the horizontal, that would be equal to 45 plus theta.
- If I draw a.

Example or application from the lecture

- That is this one, this line, is also given and that angle is forty-five degrees, which you have to note down.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000074 · EV0000075 · EV0000076 · +1 source units in pack.json

Core explanation · 05:00–09:59

The reaction, if I draw the reaction here, this total angle will be 90 degrees.

Key details

- Since this would be 45 minus theta, the angle made by the reaction at A with the horizontal will be 45 plus theta.
- And this angle is 45 plus theta, and this reaction will be at 90 degrees to this line.
- The angle made by the reaction with the horizontal will be 45 minus theta.
- We have removed the V group.
- The reactions at A, the reactions at B is indicated.
- The X and Y axis are indicated.
- You can see the sulphate of the cylinder is indicated as Mg, that is reacting vertically down.
- Next with the horizontal is going to be 45 plus theta.
- I have explained how to get that.
- I have explained that also.
- I hope it is clear till now.
- You have to identify that the reaction at A is considered as A itself, the reaction at B is considered as B.
- Which is equal to one half of that A.

- Reaction at B has to be one half of A.
- For that condition, we have to find out what is this T.
- At the center of the cylinder.
- By considering the reaction at B, this is reaction at B which is equal to one half of K.
- This is the reaction at A, which is A itself.
- This is the cell fate of the cylinder that is Mg.
- would be 45 plus theta is this minus 90 minus 45 plus theta so this would be 45 minus theta that is what is indicated here.
- Tan of 45 minus theta, which will be equal to the opposite side by adjacent side, which is B by A, and we know that B is one half of A, so it is A by 2 divided by A.
- But here instead of taking the distances, we have taken the reactions as the size.
- Or the force at the sides of the triangle, which will still work.
- You can represent the distances by forces or force by distances.
- However, if you want to solve it by using the equation of equilibrium, you can also solve it by taking.
- Sum of all the forces along X to be equal to 0.
- That means your force F cause of 45 minus Teta, which would be this, would should be equal to the reaction V cause of this.
- Cause of forty-five minus theta which is this angle.
- The value of theta.
- Along with that, we can also use some of all the forces along acting along y.

Example or application from the lecture

- Let's continue.
- That is what is given in the statement of the problem.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000078 · EV0000079 · EV0000080 · +1 source units in pack.json

Core explanation · 10:00–14:59

B sin 45 minus theta.

Key details

- You can use that also to solve this.
- Hope you have understood.
- I've considered problem 3 bar 31.
- The statement of the problem states determine the minimum cylinder mass m1 required to cause loss of contact at B.
- This mass m1 is attached to this cylinder.
- The cylinder is.
- Hanging by this row or wire.
- The wire is attached to this beam.
- One end of the wire is attached to this point beam.
- You can also see that this is an R.

- The mass of this entire.
- AOL the L angle or L member is to be considered as M.
- Whereas the mass of OB will be equal to M by 3.
- The force will always be acting vertical.
- That is 2 by 3 m times g will be acting at the center of AO, which would be at a distance of 2 by 3 L divided by 2, that would be L by 3, whereas the mass m by 3.
- Would M G by three will be acting at this point center and this distance will be equal to L by three into half or into half, which will be L by six, that distance where the force is going to be acting.
- Then you can see that here.
- The point O, there is an hinge, so there will be two reactions, one vertical and one horizontal.
- Make the contact here.
- There should not be any contact.
- So that means the reaction at A should be equal to 0.
- The free body diagram for this M1, if you take if I take the tension in this row as T.
- The same rope is continuing here, so you will have T here also.
- M1 by M1 G by 2, that is what is the force that's going to be acting here.
- This force will be acting downwards, which will make this two rotate in such a way that there is no contact here.
- So if I take some of all the vertical forces acting on the mass m1g which I have explained it just now, so I get t to be equal to m1g by 2 because you have 2t here just to be balanced by this m1g.

Example or application from the lecture

- Let's go to the next problem.
- So the condition that is given is there should not be any contact at A.
- They have given the inclination of this arm AO with respect to vertical as 30 degrees.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000082 · EV0000083 · EV0000084 · +1 source units in pack.json

Core explanation · 15:00–19:59

Then you have the sulfate of this R which is 2 mg by 3.

Key details

- Then the sulphate of OB which is mg by 3 because this length and this length are 2 L by 3 and L by 3.
- Then we also know that this length the inclined length of OA is L by 6.
- This distance is L by 6.
- This inclination is 30 degrees, so obviously this will be 60 degrees.
- This if you observe AOB is at an angle of 90 degrees.
- We know that this is 30, so if I draw a vertical line here, this would be 30, this would be 60, so this obviously has to be 30 degrees.

- Because we know that there are two reaction that is acting here, which are unknown.
- So I have to use the first equation of the equilibrium, which is the moment equation, at the point where you have the maximum or not.
- Sine 30 degrees, and that's what is mentioned here, which will produce an anti-clockwise moment, so positive.
- This would produce a clockwise moment, so negative.
- This force is mg by 3 into the horizontal distance of this force from O will be L by 3.
- Cost 30 degrees because this angle is 30 degrees.
- Then we also have this tension T which will also produce a clockwise movement.
- The horizontal distance of B from O.
- will be equal to L by 3 into cos thirty degrees and that's what is mentioned here.
- So if I simplify this I will get the value of M1 because M will in terms of M, M1 we will get it in terms of M.
- Hope everybody has understood this.
- That's why we are using only the moment equation here and we are stopping it.
- But the extension of this problem can also be if they ask you to determine the reactions are to O.
- Then if I use summation of all the forces along y to be equal to 0, so I'll get the value of OY which will be equal to m into g, summation of this, plus this T would be the reaction that's going to be acting at OY.
- Now O X is 0 because there is no.
- Horizontal force that's going to be acting here because this was anyhow zero.
- Hope everybody has understood this two problems.
- Whatever I have discussed.
- What we need to do is only use the equations of equilibrium, the three equations of equilibrium, algebraic sum of all the moments to be equal to zero at any point.
- Then summation of all the forces along y should be equal to zero.
- The algebraic sum of all the forces acting along y direction to be equal to zero, the algebraic sum of all the forces acting along x direction to be equal to zero.
- And that's what we have done now.
- But you can use only the earth equation, whatever is required for you to determine the unknown value, okay?
- So we'll stop today's class here, we'll continue in the next class.
- My name is Dr.
- See you in the next class.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000086 · EV0000087 · EV0000088 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
UE26ME142A_Unit 2_Class 18_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/018_UE26ME142A_Unit_2_Class_18_AVSummary__dafbae985b8a08a6.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 18_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/018_UE26ME142A_Unit_2_Class_18_AVSummary__dafbae985b8a08a6.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 18_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/018_UE26ME142A_Unit_2_Class_18_AVSummary__dafbae985b8a08a6.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9995; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 18_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/018_UE26ME142A_Unit_2_Class_18_AVSummary__dafbae985b8a08a6.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 2 rendered pages in 014_UE26ME142A_Unit_2_Class_14_Lecture_PPT__68b1ea1b872acf84.slides.md have no readable `.slides.md` text; their original page images are included.
2. 4 rendered pages in 015_UE26ME142A_Unit_2_Class_15_Lecture_PPT__ad0881da453d8078.slides.md have no readable `.slides.md` text; their original page images are included.
3. 2 rendered pages in 016_UE26ME142A_Unit_2_Class_16_Lecture_PPT__4b521e78b51c81b7.slides.md have no readable `.slides.md` text; their original page images are included.

4. 4 rendered pages in 017_UE26ME142A_Unit_2_Class_17_Lecture_PPT__036572c14ae14d94.slides.md have no readable `.slides.md` text; their original page images are included.
5. 3 rendered pages in 018_UE26ME142A_Unit_2_Class_18_Lecture_PPT__a852acc6b8941200.slides.md have no readable `.slides.md` text; their original page images are included.
6. 3 rendered pages in 019_UE26ME142A_Unit_2_Class_19.0_Lecture_PPT__085bbdaae12394ff.slides.md have no readable `.slides.md` text; their original page images are included.
7. 2 rendered pages in 019_UE26ME142A_Unit_2_Class_19.1_Lecture_PPT__14bcc58873e77b45.slides.md have no readable `.slides.md` text; their original page images are included.
8. 4 rendered pages in 019_UE26ME142A_Unit_2_Class_19.2_Lecture_PPT__444fedd8753b84b9.slides.md have no readable `.slides.md` text; their original page images are included.
9. 3 rendered pages in 019_UE26ME142A_Unit_2_Class_19.3_Lecture_PPT__7981db4bc773065f.slides.md have no readable `.slides.md` text; their original page images are included.
10. 2 rendered pages in 019_UE26ME142A_Unit_2_Class_19.4_Lecture_PPT__54dd570fb45dd727.slides.md have no readable `.slides.md` text; their original page images are included.
11. 2 rendered pages in 019_UE26ME142A_Unit_2_Class_19.5_Lecture_PPT__c337d52a52027bf0.slides.md have no readable `.slides.md` text; their original page images are included.
12. 3 rendered pages in 021_UE26ME142A_Unit_2_Class_21_Lecture_PPT__66c97ce40e0b782a.slides.md have no readable `.slides.md` text; their original page images are included.
13. 2 rendered pages in 022_UE26ME142A_Unit_2_Class_22_Lecture_PPT__b1114b6d964d54c7.slides.md have no readable `.slides.md` text; their original page images are included.
14. 3 rendered pages in 023_UE26ME142A_Unit_2_Class_23_Lecture_PPT__433882c2bbb0b6a0.slides.md have no readable `.slides.md` text; their original page images are included.
15. 2 rendered pages in 024_UE26ME142A_Unit_2_Class_24_Lecture_PPT__c0fa0555c4f1df8b.slides.md have no readable `.slides.md` text; their original page images are included.
16. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
17. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.

18. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
19. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
20. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00018	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0054. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00019	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0054. Sidecar variants compare...
SRC00020	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0054. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00021	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0054. Sidecar variants compare...

C Revision and retrieval practice